

循序渐进，学习开发 xv6-riscv

第 4 章 物理内存管理

汪辰



目录

01

物理内存页的组织

02

物理内存页的分配与释放

03

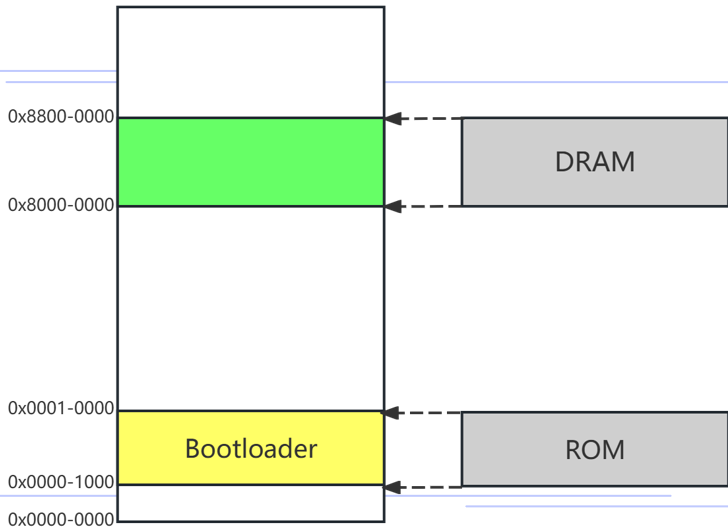
`mem*/string` 函数



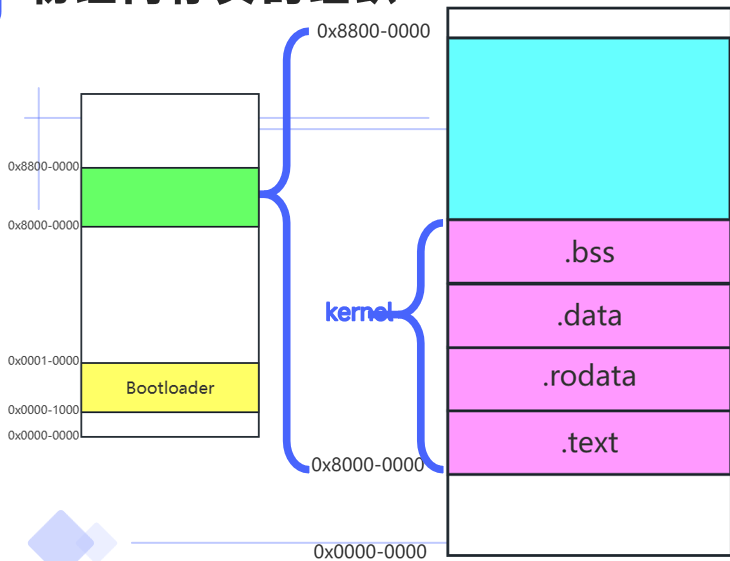
01

物理内存页的组织

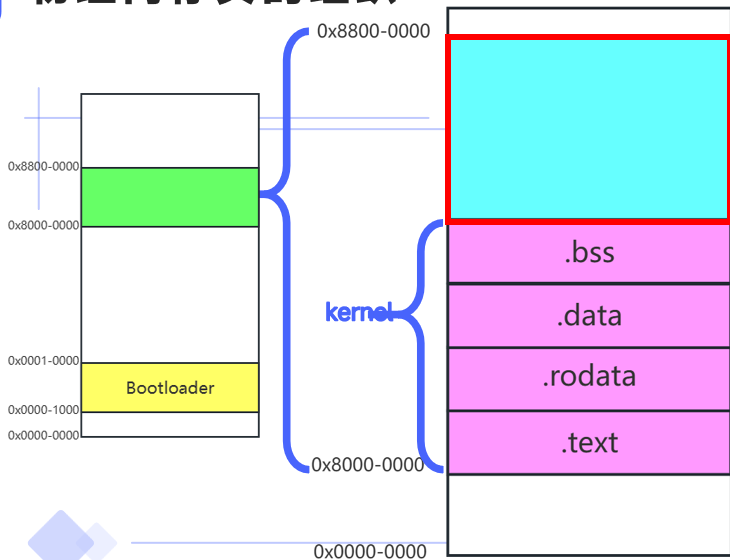
物理内存页的组织



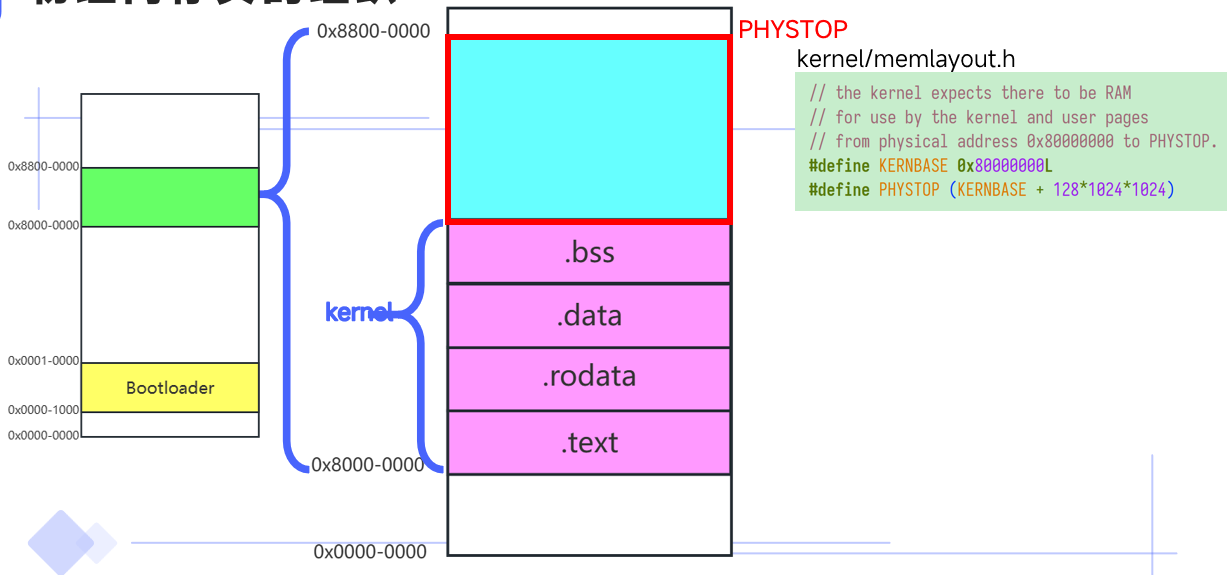
物理内存页的组织



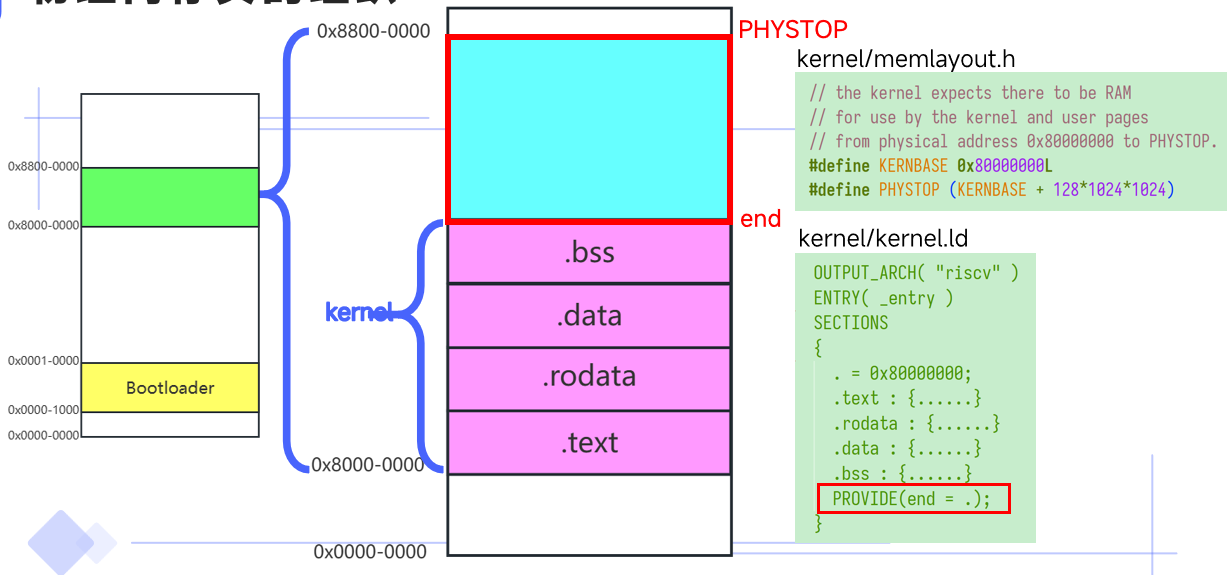
物理内存页的组织



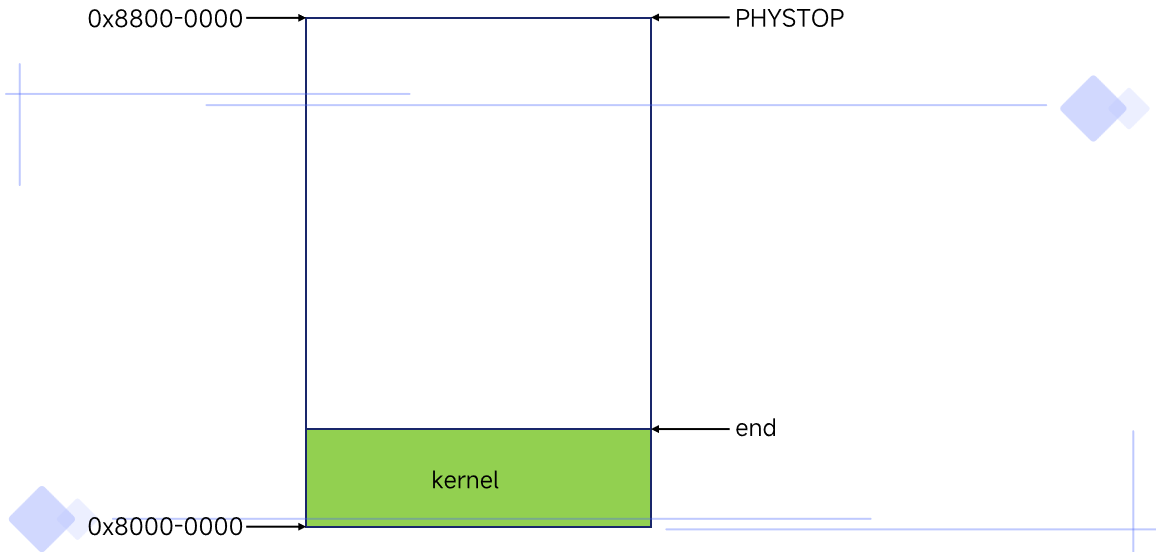
物理内存页的组织



物理内存页的组织



物理内存页的组织



物理内存页的组织

0x8800-0000

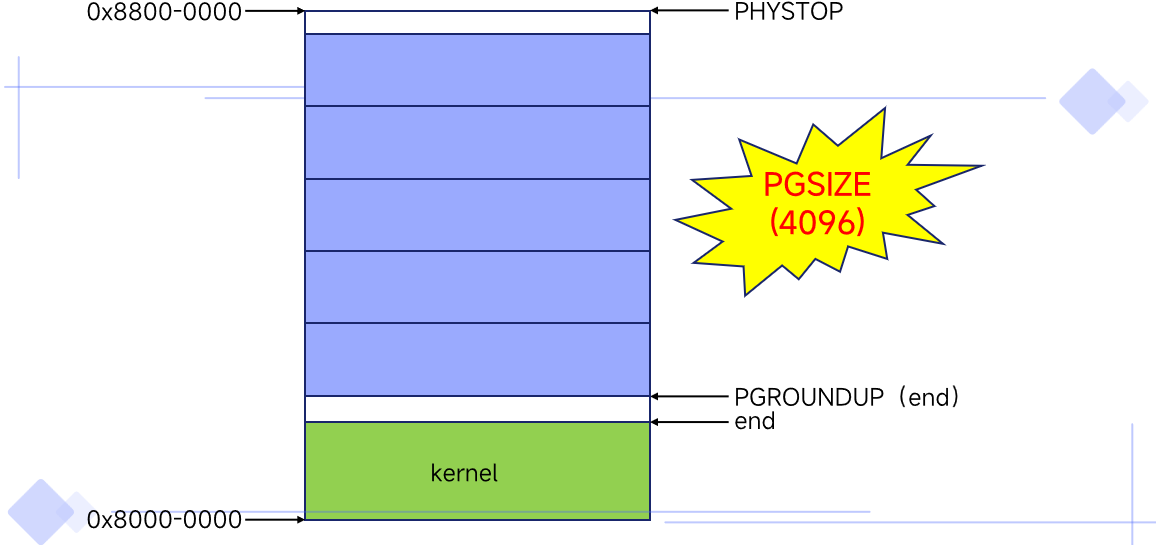
PHYSTOP

PGSIZE
(4096)

PGROUNDUP (end)
end

kernel

0x8000-0000



物理内存页的组织

0x8800-0000

PHYSTOP

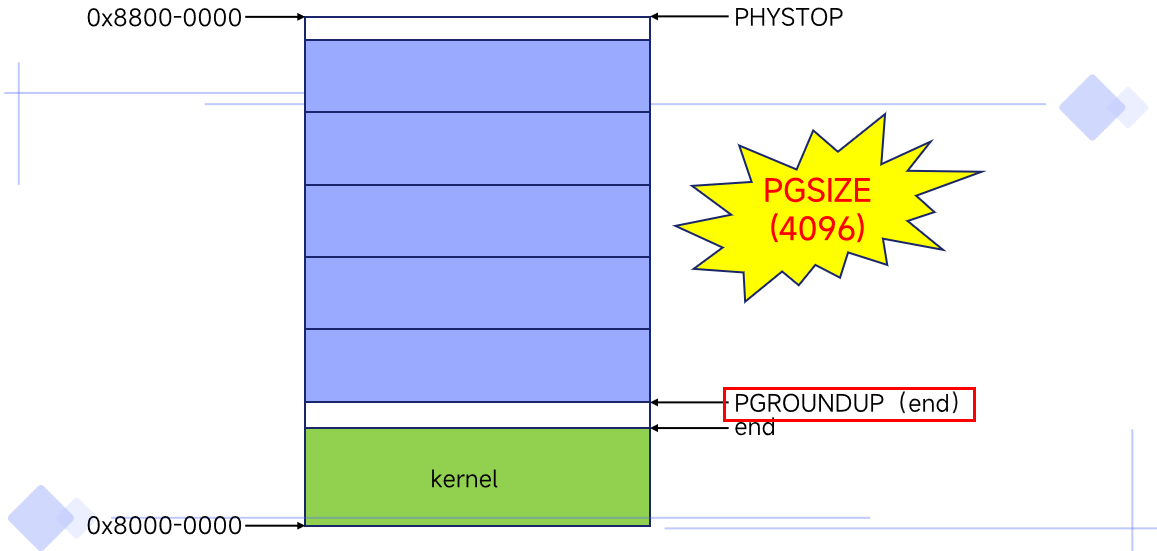
PGSIZE
(4096)

PGROUNDUP (end)

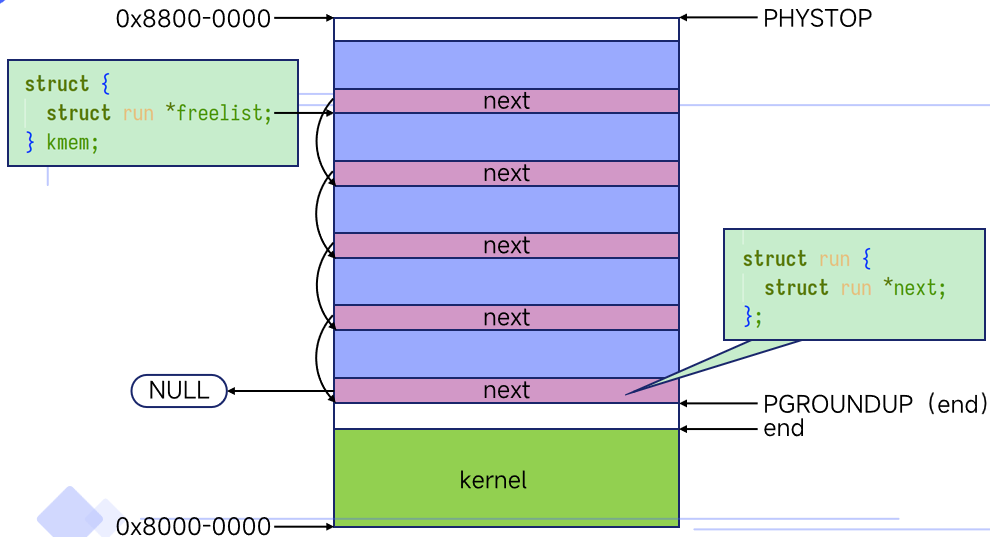
end

kernel

0x8000-0000



物理内存页的组织



A decorative graphic on the left side of the slide. It features two concentric circles with a light blue glow. A thin grey arc starts from the top of the outer circle and curves around to the bottom. In the top right corner, there is a solid blue rectangle.

02

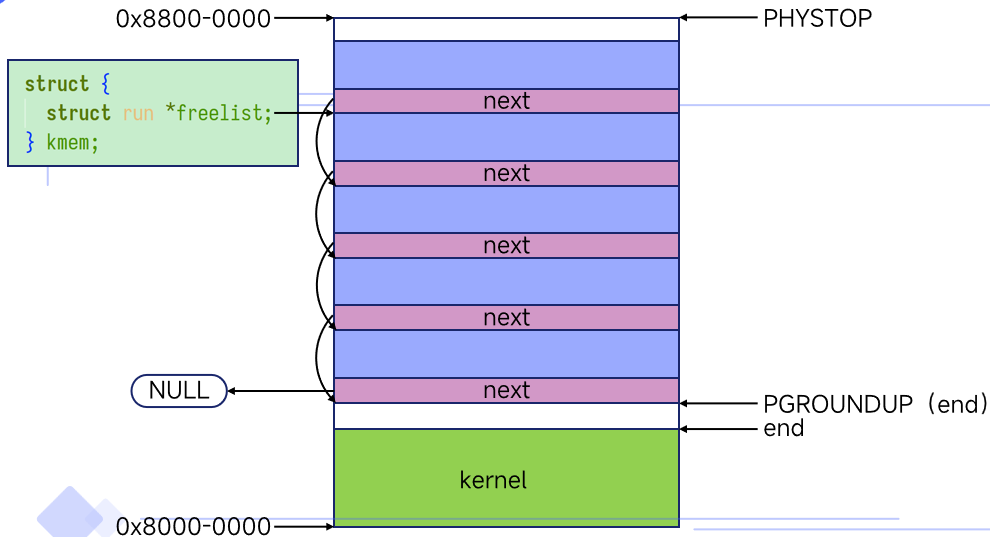
物理内存页的分配与释放

物理内存页的分配与释放 - API

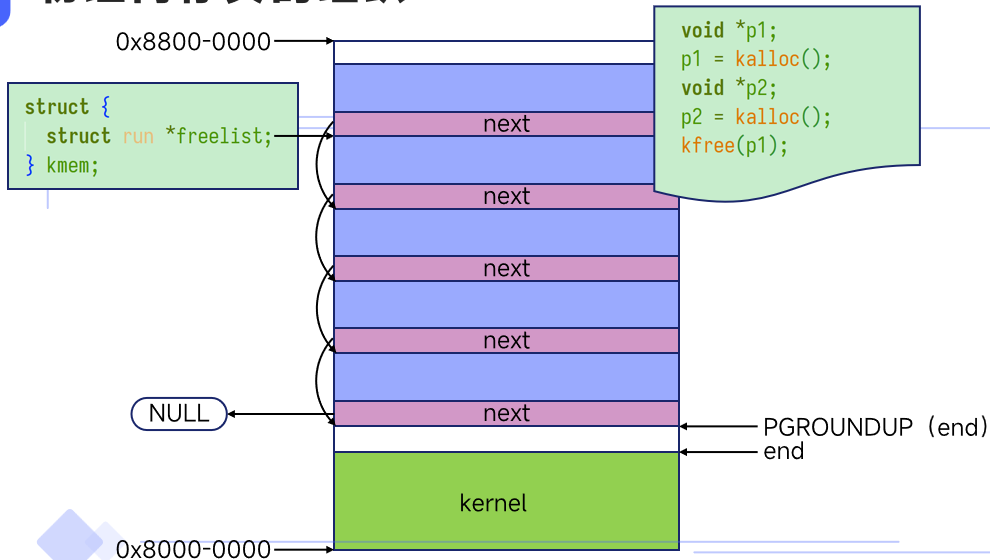
```
// kalloc.c
```

```
void*      kalloc(void);  
void       kfree(void *);  
void       kinit(void);
```

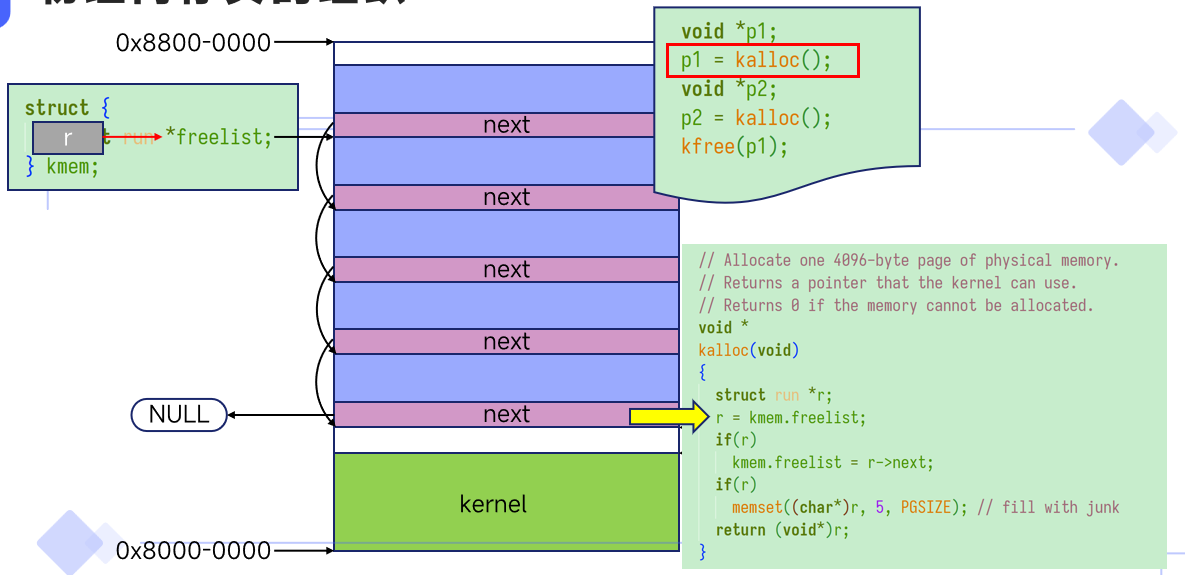
物理内存页的组织



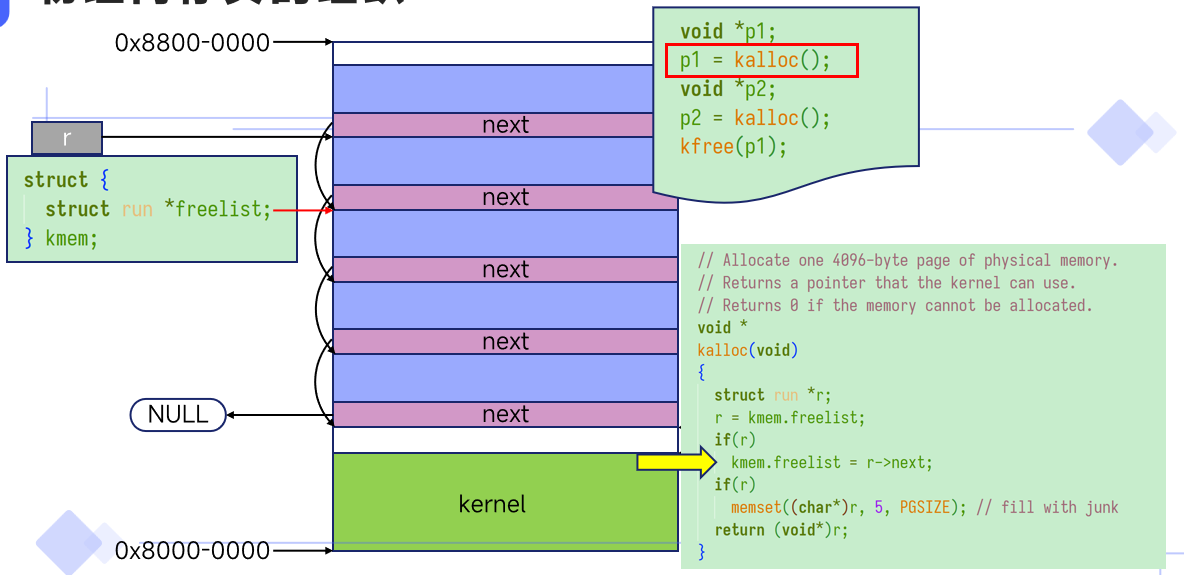
物理内存页的组织



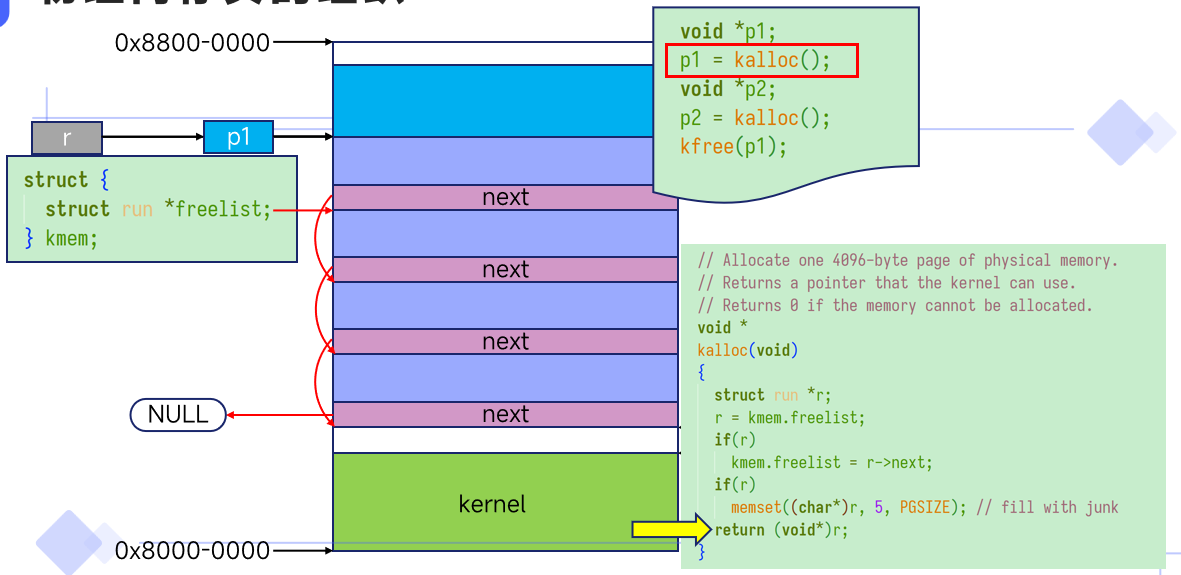
物理内存页的组织



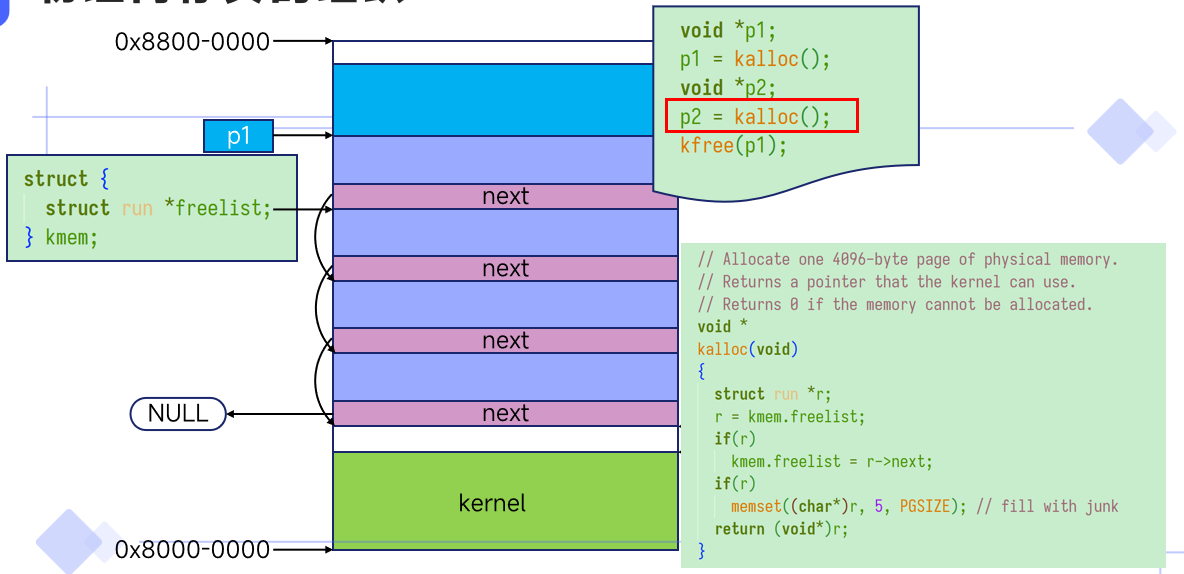
物理内存页的组织



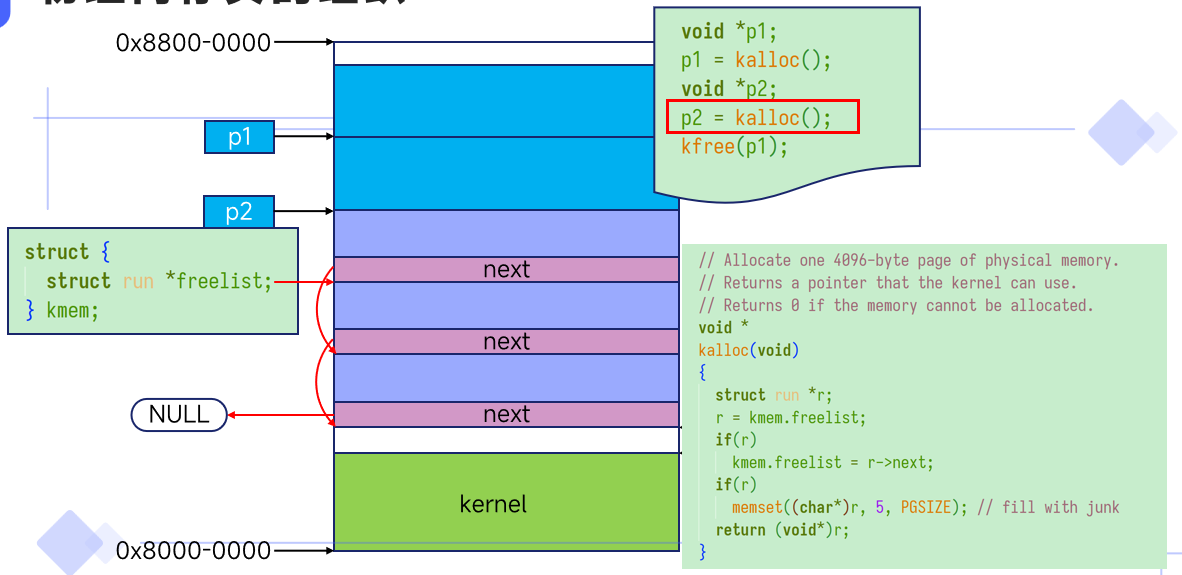
物理内存页的组织



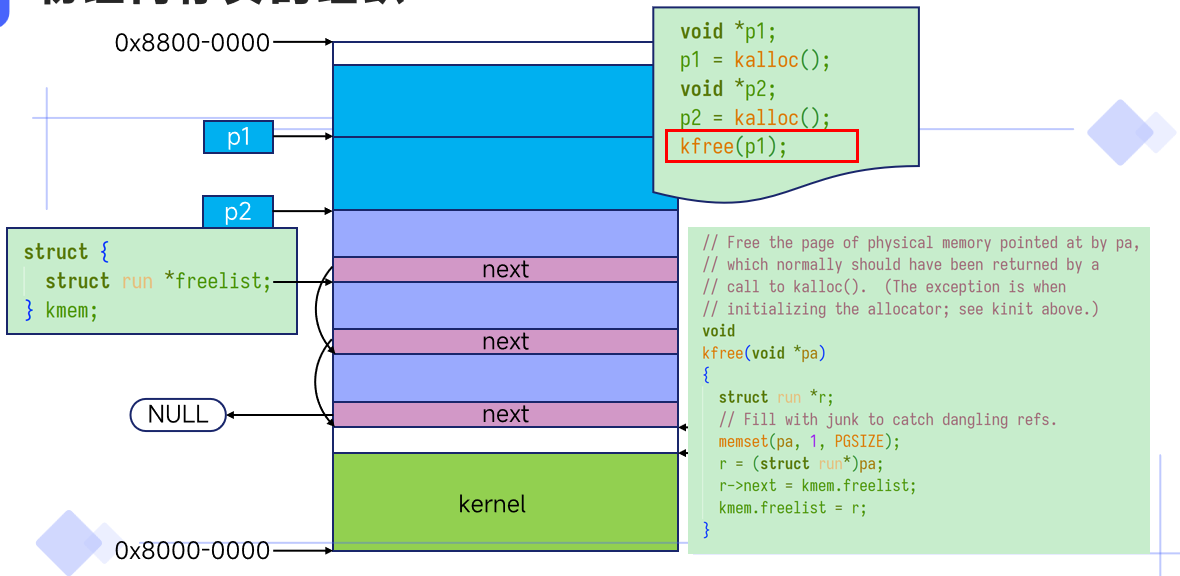
物理内存页的组织



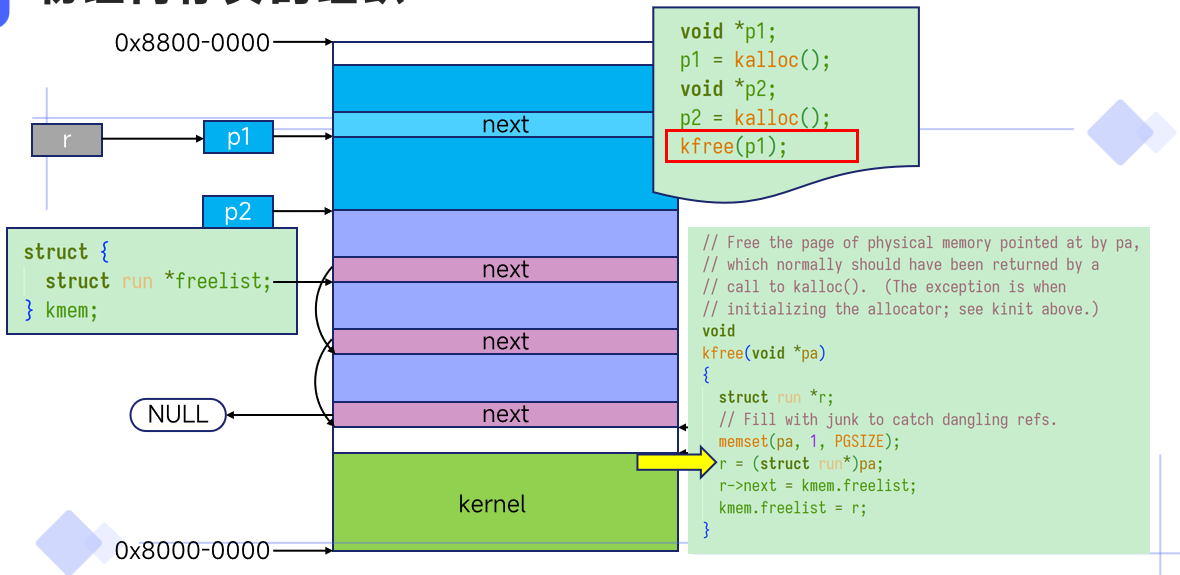
物理内存页的组织



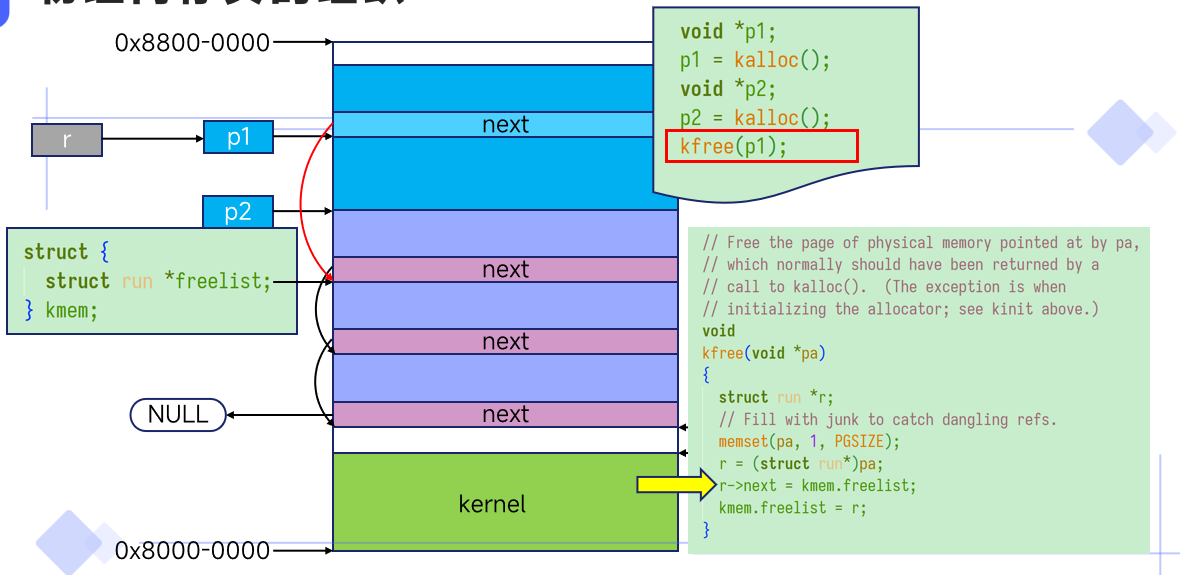
物理内存页的组织



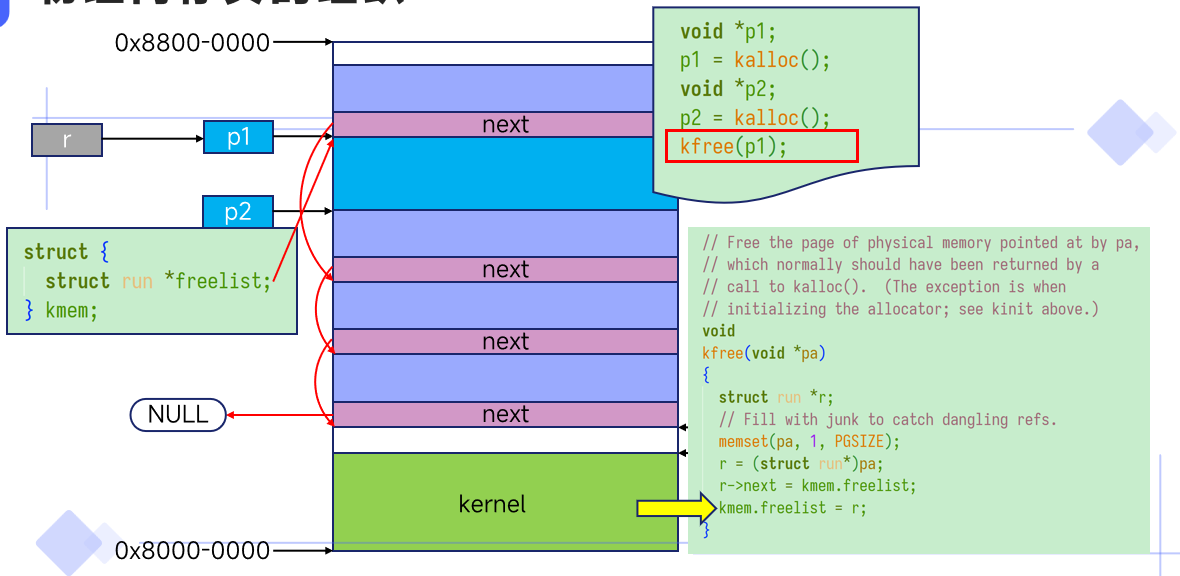
物理内存页的组织



物理内存页的组织



物理内存页的组织





03

mem*/string 函数

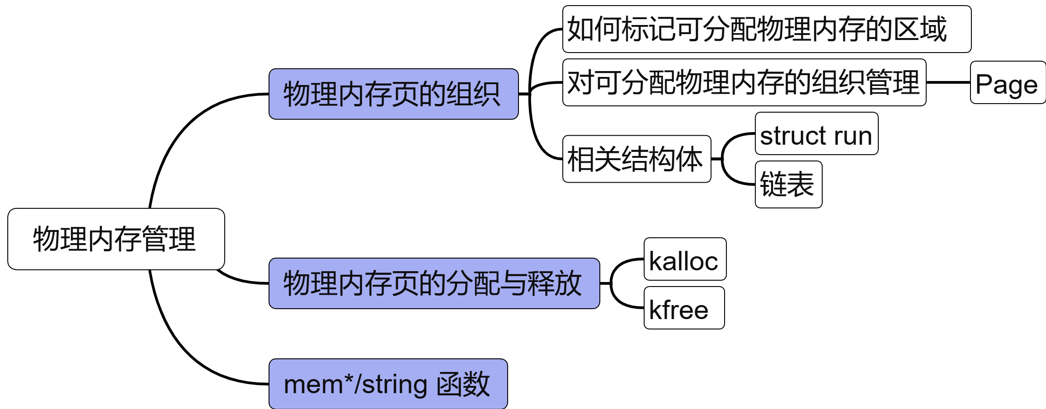
mem*/string 函数

// string.c

```
int      memcmp(const void*, const void*, uint);
void*    memmove(void*, const void*, uint);
void*    memset(void*, int, uint);
char*    safestrcpy(char*, const char*, int);
int      strlen(const char*);
int      strncmp(const char*, const char*, uint);
char*    strncpy(char*, const char*, int);
```



本章总结



谢谢

